Ali Nikkhah

Tehran, Iran | Open to Relocation | alinkkh9@gmail.com | +98 991 296 3951
alinikkhah2001.github.io | linkedin.com/in/alinikkhah2001

Research Interests

Vision-language models; multimodal representation learning; explainable and robust AI; speech and audio intelligence; applied machine learning for healthcare and financial systems.

Education

Sharif University of Technology Tehran, Iran
B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major *Sept 2020 – Jul 2024*

- Relevant coursework: Deep Learning, Machine Learning, Digital Signal Processing, Linear Algebra, Probability & Statistics, Algorithm Design, Digital Systems Design.

Publications & Preprints

Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning 2025
Ali Nikkhah, [Co-authors TBD], **A. Supervisor**
arXiv preprint – in preparation

Research Experience

University of British Columbia Remote
Research Assistant – Vision-Language Medical Imaging *Feb 2024 – Nov 2025*

- Developed vision-language architectures (**ViT**, **BLIP**, **T5**) trained with self-supervised contrastive learning to generate DICOM-compliant diagnostic reports from raw ultrasound scans; achieved SOTA **BLEU** and **ROUGE-L** on the evaluation benchmark.
- Investigated cross-modal alignment strategies to bridge the semantic gap between ultrasound image features and structured radiology report language.
- Work resulted in a preprint in preparation (see Publications).

Trinity College Dublin – EmoDub Project Remote
ML Researcher – Emotion-Aware Voice Translation *May 2025 – Nov 2025*

- Researched emotion-preserving voice translation, developing architectures that maintain both linguistic content and affective characteristics across language boundaries.
- Fused acoustic, semantic, and contextual embeddings using **wav2vec2.0** and **Whisper** to represent speaker emotion independently of lexical content, enabling emotionally faithful dubbing generation.

L3S Research Center (Leibniz University Hannover) Remote
Research Assistant – Video Motion Classification *Apr 2023 – Feb 2024*

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders applied to dense optical flow fields, deliberately suppressing appearance cues to isolate generalizable human-dynamic representations.
- Achieved state-of-the-art classification accuracy on **UCF-101** and **HMDB-51** benchmarks; findings demonstrated that flow-based texture-agnostic features transfer more robustly across recording conditions.

Sharif University of Technology – Applied ML Lab Tehran, Iran
Research Collaborator – Explainable AI *Jul 2025 – Sep 2025*

- Investigated domain-agnostic Explainable AI (XAI) frameworks, applying advanced attention visualization techniques to probe the internal representations of large neural networks.
- Explored robustness of ML model explanations under distribution shift, contributing to lab discussions on interpretability evaluation methodology.

Teaching & Mentoring

Sharif University of Technology Tehran, Iran
Graduate & Undergraduate Teaching Assistant *Jan 2021 – Jan 2024*

- **Foregrounded courses (graduate-level):** Deep Learning, Natural Language Processing, Data Science, Machine Learning.
- **Additional courses (undergraduate-level):** Generative Models, Engineering Probability & Statistics, Analog Electronics, Circuit Theory.
- Responsibilities across roles included designing and grading problem sets, holding weekly office hours for cohorts of 30–80 students, preparing tutorial sessions, and contributing to exam question design for graduate-level deep learning and NLP courses.
- 3 years continuous TA service across 8 distinct courses spanning both core electrical engineering and advanced AI/ML curricula.

Industry Experience

Digikala Tehran, Iran
AI Engineering R&D Lead *Nov 2024 – Aug 2025*

- Led applied research into agentic RAG architectures, implementing Knowledge-Augmented Generation (KAG), hybrid dense/sparse retrieval (RRF), and HyDE query expansion in a production system serving 5M+ daily queries.
- Investigated multi-agent system design using LangGraph: modeled tool-calling agents as cyclic directed graphs with explicit state transitions, evaluating tradeoffs between agent autonomy and output reliability.

Advanced Analytics Australia Remote Contract
Computer Vision Engineer *Sep 2025 – Apr 2026*

- Applied deep vision models (ViT, object detection, optical flow) to real-world safety compliance; investigated quantization and runtime compilation (TensorRT, TF Lite) for constrained edge deployment.

Turquoise Digital (Firouzeh Digital) Tehran, Iran
Senior Data Scientist & Data Engineer *Aug 2025 – Mar 2026*

- Applied time-series transformer models to financial forecasting; designed agentic chatbot with function-calling for secure live data retrieval; built telephony audio transcription pipeline using Whisper and wav2vec2.0.

HomaCloud Tehran, Iran
MLOps Engineer *May 2021 – Feb 2022*

- Orchestrated distributed ML pipelines on Kubernetes (Ray, MLflow, TensorFlow Serving); managed CI/CD and infrastructure-as-code (Terraform, ArgoCD).

Technical Skills

Languages: Python, C++, Go, SQL, R, Bash, TypeScript
Deep Learning & Vision: PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE, KAG, Graph RAG
Data & MLOps: Spark, Airflow, Kafka, Docker, Kubernetes, Ray, Triton, vLLM, MLflow, W&B
Languages (Human): Persian (Native), English (C2 – IELTS 8.0), Arabic (Professional), French (Beginner)